

BdPhO-2021-2022
National Round, Category C

Write all your final answers in the empty spaces provided and do the rough work on the opposite side the pages that are empty. If you need extra pages ask your invigilator.

Vagabond on a Plane

A point with mass m is moving in a 2 dimensional plane. The motion can be described either in terms of a pair of Cartesian coordinates (x, y) or in terms of polar coordinates (r, θ) . Here r is the distance of the mass measured from the (fixed) origin and θ is the angle measured with respect to the x -axis.

(a) Write the transformation between the x, y coordinates and the r, θ coordinates. **1**

(b) Given your answer to (a), how can one obtain the components of the velocity of the particle along the radius (radial velocity) as well as the component along the tangential direction (along the direction of increasing value of θ). **1.5**

(c) Therefore, express the kinetic energy of the particle in terms of these velocity components which you found in (b). **1**

(d) What is the angular momentum of this particle when the particle is at a distance r from the center and time derivatives of the polar coordinates are known to be $(\dot{r}, \dot{\theta})$. **0.5**

(e) Reexpress the kinetic energy but including the angular momentum which is denoted by l hereafter. **1**

(f) Let the potential energy of the particle V be a function of the distance , i.e. $V = V(r)$. If the total energy of the particle is E , what is the kinetic energy that is caused by the motion in the radial direction in terms of the quantities appearing in (e)? **1**

(g) As the potential energy is independent of θ (as in this case), this problem can be treated as an one dimensional motion - occuring along the radial direction alone. That's why your answer in the above must only involve the coordinate r and its derivatives. What is the effective potential energy for the system that you can find from your answer to (f)? **1**

(h) Let us take the case where $V(r) = -\frac{K}{r^2}$. Draw two sketches of the effective potential energy as a function of r alone describing two cases, namely, (i) When the angular momentum $l \gg l_0$ (i.e. l is very large compared to certain fixed l_0), (ii) and $l \ll l_0$. Assume $K > 0$. **1+1**

(i) Determine the value of l_0 in terms of the other constants that we have described in the previous parts of the problem **1.**

A charge who became Narcissus

Consider a particle carrying a charge $+Q$ ($Q > 0$) which is placed at a distance D away from an infinite metallic plate in a state of rest. The plate is grounded.

(a) What is the electric potential Φ of the plates? **0.5**

(b) Draw the electric field lines between the plate and the point charge and also label the direction of the field lines. **1+ 0.5**

(c) If the coordinate is such that the charge in part (a) has the (Cartesian) coordinate $(D, 0)$, What is the electrostatic potential at a point (x, y) with $x > 0$? **3**

(d) What is the value of the electric field *on the surface of the metallic plate* if the distance between given charge and point of observation, is denoted by $r > d$? **2**

(e) What is the surface charge density induced on the plate as a function of this distance r ? If an observer see the system from very far but staying on the plate how does the charge appear to the observer? **1+2**